

Aditya Vasu Rana

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Education

Vellore Institute of Technology, Chennai

Aug 2023 – May 2027

Bachelor of Technology (B.Tech) in Computer Science, CGPA : 9.5 (Branch Rank 3)

- **Coursework:** Computer Networks, Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Database Management Systems

Technical Skills

- **Programming Languages:** Python, R, C, C++ ,Java
- **Data Analysis & Modeling:** Machine Learning, Neural Networks, Transformers
- **Tools & Platforms:** Git, AWS, Docker, Jupyter, Streamlit , Hex
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn
- **Databases & Visualization:** SQL, PL/SQL, Tableau, Power BI, Data Storytelling
- **Web Technologies:** HTML, CSS, JavaScript, ReactJS, Flask, FastAPI, Tailwind, Bootstrap

Projects

Fashion Recommender System - CNN + ResNet50 + KNN + FastAPI Backend [↗](#)

- Built a **content-based fashion recommendation system** using **ResNet50 (ImageNet pretrained)** as a feature extractor to generate **2048-D deep embeddings** for 44k fashion product images.
- Implemented an efficient image retrieval pipeline using **GlobalMaxPooling**, **L2 normalization**, and **K-Nearest Neighbors (KNN)** to recommend the **Top-5 visually similar items**.
- Developed an interactive **Python FastAPI web application** that allows users to upload a fashion image and receive real-time recommendations using precomputed embeddings stored in `embeddings.pkl`.

AI-Powered Applicant Tracking System (ATS) - FastAPI + React.js + PostgreSQL + NLP [↗](#)

- Built a **full-stack recruitment platform** with **FastAPI**, **React.js**, **PostgreSQL**, **SQLAlchemy**, and **JWT authentication**, enabling role-based workflows for recruiters and candidates.
- Developed an automated **resume-job matching pipeline** using **PyMuPDF** and NLP techniques to extract skills, generate **ATS scores**, and compute semantic similarity between resumes and job descriptions.
- Implemented application tracking, candidate ranking, shortlisting/rejection workflows, recruiter feedback, and interview question generation to streamline the hiring process.

Customer Churn Prediction - Machine Learning + Flask Backend [↗](#)

- Built an end-to-end customer churn prediction system using Logistic Regression and Random Forest on telecom customer data, achieving up to **0.78 accuracy** and **0.72 recall for churn class**.
- Implemented a robust ML pipeline with EDA, preprocessing, class-imbalance handling (SMOTE and class weighting), and threshold tuning to improve minority-class detection.
- Developed an interactive **Flask-based web application** with SHAP-powered explainability to surface top risk drivers and enable real-time churn risk assessment with dynamic churn probability threshold tuning.

Experience

Data Science and Analytics (D2C Insights) Intern, Kerala Ayurveda Ltd.

Oct 2025 – Jan 2026

- Analyzed D2C sales funnel across multiple product categories using Python and SQL, identifying key drop-offs such as low Add-to-Cart (ATC) and Cart-to-Checkout (C2C) rates for sales improvement initiatives.
- Conducted customer behavior analysis including low ATC diagnosis, abandoned cart AOV analysis, and first-product to next-likely-product mapping, generating actionable insights to optimize conversion journeys.
- Collaborated closely with the product management team to frame hypotheses, validate marketing strategies, and support data-driven decisions aimed at improving sales performance and customer engagement.

Supply Chain Analytics Intern, Chambal Fertilisers and Chemicals Ltd.

May 2025 – Jul 2025

- Analyzed fertilizer distribution and logistics data using Python and SQL to identify transportation bottlenecks, optimize route efficiency, and support cost reduction initiatives across regional supply networks.
- Developed demand forecasting models and inventory planning reports by leveraging historical sales and dispatch data, enabling improved stock allocation and regional inventory management.
- Created operational dashboards and KPI reports to monitor warehouse performance, inventory turnover, and supply chain efficiency, providing data-driven insights to support business decision-making.

Certifications and Achievements

- **Oracle Cloud Infrastructure Architecture: Data Science Professional**, Oracle University [↗](#)
- NDA AIR 4 (2022-1) – Secured All India Rank 4 in NDA entrance exam.
- 2nd place at V-Medithon Intercollege Hackathon, Ranked 1 for ML model.
- 300+ problems solved on LeetCode with 100 Days Badge [↗](#)